

Functions

1) $y = 3x + 5$



2) $2x + 3$

3) $g(x) = 4 - x$

4) $f(x) = -\frac{2}{3}x + 5$

1) $f(x) = -3x^2 + 2x - 4$

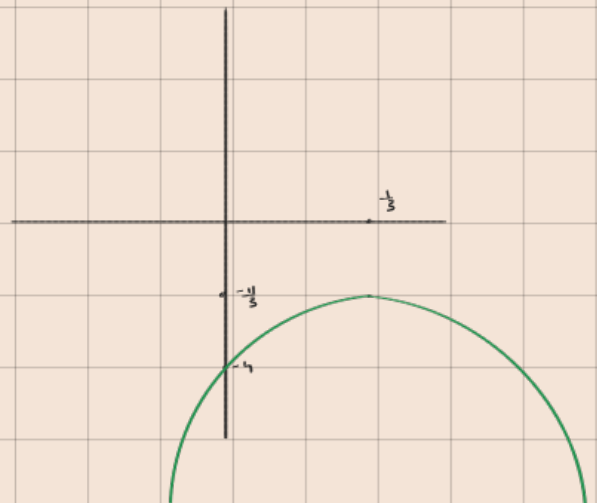
$$x = \frac{-(-2)}{2(-3)}$$

$$x = \frac{1}{3}, -\frac{11}{3}$$

$$\text{vertex} \left(\frac{1}{3}, -\frac{11}{3} \right)$$

$$\text{domain} = (-\infty, \infty)$$

$$\text{range} = (-\infty, -\frac{11}{3}] \cup [-\frac{11}{3}, \infty)$$



2) $6(x+1)^2 - 11$

$$= 6(x^2 + 2x + 1) - 11$$

$$= 6x^2 + 12x + 6 - 11$$

$$= 6x^2 + 12x - 5$$

$$\text{vertex} = (-1, 11)$$

$$x = \frac{-(-12)}{2(6)} \quad y = -11$$

$$= \frac{-12}{12}$$

$$= -1$$

$$\text{Domain} = (-\infty, \infty)$$

$$\text{Range} = (-\infty, -1] \cup [-1, \infty)$$

limits

$$1) \quad \lim_{n \rightarrow \infty}$$

$$\frac{\sin ax}{\sin bx}$$

$$= \frac{\sin ax}{\sin bx} \times \frac{abx}{abx}$$

$$= \frac{\sin ax}{abx} \div \frac{\sin bx}{abx} \quad \frac{\sin ax}{ax} = 1$$

$$= \frac{1}{b} \div \frac{1}{a}$$

$$= \frac{a}{b}$$

$$2) \quad \lim_{x \rightarrow 0} x \cdot \sin\left(\frac{1}{x}\right) \quad ?$$

$$3) \quad \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n$$

$$= \left(1 - \frac{1}{n}\right)^n$$

$$= \left(\frac{n-1}{n}\right)^n$$

$$= \left(\frac{n-1}{n-1}\right)^{-n}$$

$$= \left(\frac{n-1}{n-1} + \frac{1}{n-1}\right)^{-n}$$

$$= \left(1 + \frac{1}{n-1}\right)^{-n}$$

$$= \left(1 + \frac{1}{n-1}\right)^{-n}$$

$$= \frac{1}{\left(1 + \frac{1}{n-1}\right)^{n-1} \left(1 + \frac{1}{n-1}\right)^1}$$

$$= \frac{1}{e \cdot 1}$$

$$e = \left(1 + \frac{1}{n}\right)^n$$

$$= \frac{1}{e \cdot 1}$$

$$= \frac{1}{e}$$

$$\left[\left(1 - \frac{1}{n}\right)^{-n}\right]^{-1}$$

$$\left[\left(1 + \frac{1}{n-1}\right)^{-n}\right]^{-1}$$

$$= e^{-1}$$

$$4) \quad \lim_{n \rightarrow \infty} \left(\frac{n}{1+n} \right)^n$$

$$= \left(\frac{n+1-1}{1+n} \right)^n$$

$$= \left(\frac{1+n}{1+n} - \frac{1}{1+n} \right)^n$$

$$= \left(1 - \frac{1}{1+n} \right)^n$$

$$= \left(1 - \frac{1}{1+n} \right)^{n+1} \cdot \left(1 - \frac{1}{1+n} \right)^{-1}$$

$$= \left[\left(1 + \frac{1}{-(1+n)} \right)^{-(1+n)} \right]^{-1} \cdot (1-e)^{-1}$$

$$= e^{-1} \cdot 1$$

$$= \frac{1}{e}$$

$$5) \quad \lim_{n \rightarrow \infty} \frac{n^4 - 2n^2 + 6}{n^2 + 7}$$

$$n^4 \left(1 - \frac{2}{n^2} + \frac{6}{n^4} \right)$$

$$\frac{1}{2} = 0$$

$$n^2 \left(\frac{n^2}{n^2} + \frac{7}{n^2} \right)$$

$$= \frac{1}{0}$$

$$= \infty$$

6)

$$x - \sqrt{x^2 - a^2}$$

$$x - \sqrt{x^2 - a^2} \cdot \frac{x + \sqrt{x^2 - a^2}}{x + \sqrt{x^2 - a^2}}$$

$$= \frac{(x - \sqrt{x^2 - a^2})(x + \sqrt{x^2 - a^2})}{x + \sqrt{x^2 - a^2}}$$

$$= \frac{x^2 - (x^2 - a^2)}{x + \sqrt{x^2 - a^2}}$$

$$= \frac{x^2 - x^2 - a^2}{x + \sqrt{x^2 - a^2}}$$

$$= \frac{-a^2}{x + \sqrt{x^2 - a^2}} \rightarrow 0$$

7)

$$\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^{\frac{1}{2}}}$$

$$\frac{x^2}{x^{\frac{1}{2}}} + \frac{1}{x^{\frac{1}{2}}}$$

$$\sqrt{x} + \frac{1}{\sqrt{x}}$$

$$= \infty + 0$$

$$= \infty$$

8)

$$\frac{5x^3 + 3x^2 - 1}{x - 4x^4}$$

$$= \frac{5x^3}{x^4} < \frac{5x^3}{x^4}$$

$$= \frac{0}{1}$$

$$= 0$$

$$9) \lim_{x \rightarrow 2} \frac{\sqrt{4-x^2}}{\sqrt{6-5x+x^2}}$$

$$\frac{4-x^2}{6-5x+x^2}$$

$$= \frac{2^2-x^2}{x^2-5x+6}$$

$$= \frac{(2+x)(2-x)}{x^2-(2+3)x+6}$$

$$= \frac{(2+x)(2-x)}{x^2-2x-3x+6}$$

$$x^2-2x-3x+6$$

$$= \frac{(2+x)(2-x)}{x(x-2)-3(x-2)}$$

$$x(x-2)-3(x-2)$$

$$= \frac{(2+x)(2-x)}{(x-3)(2-x)}$$

$$(x-3)(2-x)$$

$$= \frac{(2-x)}{x-3}$$

=

$$10) \lim_{x \rightarrow \infty} \frac{x + \sin x}{x}$$

$$\frac{x}{x} + \frac{\sin x}{x}$$

$$= 1 + 1 = 2$$

$$11) f(x) = \begin{cases} x^2-1 & x \leq 2 \\ \sqrt{x+3} & x > 2 \end{cases}$$

$$\lim_{x \rightarrow 2}$$

left

$$x^2-1$$

$$4-1 = 3$$

right

$$\sqrt{2+3}$$

$$= 3$$

limit approaches 3

$$a) \quad f(x) = \begin{cases} \cos x & x \leq 0 \\ 1-x & x > 0 \end{cases}$$

$$\lim_{x \rightarrow 0} f(x) \quad \begin{array}{ll} \text{left} & \text{right} \end{array} \quad \text{limit approaches } 1$$

$$= \cos(0) \quad 1-0$$

$$= 1 \quad = 1$$

$$b) \quad \lim_{x \rightarrow 0^-} \frac{x}{x - \ln(x)}$$

$$= \frac{x}{x - (-\ln(x))}$$

$$= \frac{x}{x + \ln(x)}$$

$$= \frac{x}{\frac{x}{2x}}$$

$$= \frac{1}{2}$$

$$c) \quad \lim_{x \rightarrow 1} \frac{x^3 + x^2 - 5x + 3}{x^2 - 3x + 2}$$

?

$$d) \quad g(t) = \begin{cases} t-2 \end{cases}$$

Continuity

1) Discuss the continuity of $x - |x|$ at $x = 1$

?

2) Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$

defined by

$$f(x) = \begin{cases} x & \text{if } x \text{ is irrational} \\ 1-x & \text{if } x \text{ is rational} \end{cases}$$

$$x = 1-x$$

$$x + x = 1$$

$$2x = 1$$

$$x = \frac{1}{2}$$

L'Hopital

$$1) \lim_{x \rightarrow \pi} \frac{\sin^2 x}{\cos 3x + 1} = \frac{0}{-1 + 1} = \frac{0}{0}$$

$$\frac{dy}{dx} = \frac{2 \sin x \cos x}{-3 \sin 3x + 0} = \frac{0}{0}$$

$$\frac{d^2 y}{dx^2} = \frac{2(\sin x \cdot \sin x + \cos x \cdot \cos x)}{-3 + 3 \cos 3x}$$

$$= \frac{-2 \sin^2 x + 2 \cos^2 x}{-9 \cos 3x}$$

$$= \frac{-0 + 2(-1)^2}{-9(-1)}$$

$$= \frac{2}{9}$$

$$2) \lim_{x \rightarrow 0} \frac{\ln(1-x^2)}{\ln(\cos x)} = \frac{0}{0}$$

$$\frac{dy}{dx} = \ln(1-x^2) \div \ln \cos x$$

$$= \frac{1}{1-x^2} \times (0-2x) \div \frac{1}{\cos x} \times -\sin x$$

$$= \frac{-2x}{1-x^2} \div \frac{-\sin x}{\cos x}$$

$$= \frac{-2x}{1-x^2} \div -\tan x$$

$$= \frac{2x}{(1-x^2)\tan x}$$

$$= \frac{2x}{\tan x - x^2 \tan x}$$

$$= \frac{2}{\sec^2 x - (x^2 \sec^2 x + \tan x \cdot 2x)}$$

$$= \frac{2}{\sec^2 x - x^2 \sec^2 x + 2x \cdot \tan x}$$

$$= \frac{2}{1 - 0 + 0}$$

$$= 2$$

$$4) \lim_{x \rightarrow 0} \frac{\cosh x - \cos x}{x \cdot \sin x}$$

$$\frac{dy}{dx} = \frac{\sinh x - (-\sin x)}{x \cdot \cos x + \sin x}$$

$$= \frac{\sinh x + \sin x}{x \cdot \cos x + \sin x}$$

$$= \frac{\cosh x + \cos x}{-x \cdot \sin x + \cos x + \cos x}$$

$$= \frac{1 + 1}{0 + 1 + 1}$$

$$= \frac{2}{2}$$

$$= 1$$

$$5) \lim_{x \rightarrow 1} \frac{1 - x + \ln x}{1 - \sqrt{2x - x^2}} = \frac{1 - 1 + 0}{1 - 1} = \frac{0}{0}$$

$$= \frac{1 - x + \ln x}{1 - (2x - x^2)^{\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{0 - 1 + \frac{1}{x}}{0 - \frac{1}{2} (2x - x^2)^{-\frac{1}{2}} (2 - 2x)}$$

$$= \frac{-1 + \frac{1}{x}}{\frac{1}{2} (2x - x^2)^{\frac{1}{2}}}$$

$$= \frac{-x + 1}{x} \div \frac{-\cancel{1} (1 - x)}{\cancel{2} (2x - x^2)^{\frac{1}{2}}}$$

$$= \frac{1 - \cancel{x}}{x} \div \frac{-\cancel{1} (1 - \cancel{x})}{\sqrt{2x - x^2}}$$

$$= \frac{1}{x} \times \sqrt{2x - x^2}$$

$$= \frac{\sqrt{2x - x^2}}{x}$$

$$= 1$$

$$b) \lim_{x \rightarrow 0} \frac{x \cdot \cos x - \ln(1+x)}{x^2}$$

$$= \frac{x \cdot \cos x - \ln(1+x)}{x^2} \quad \div \quad x^2$$

$$\frac{dy}{dx} = x \cdot (-\sin x) + \cos x - \frac{1}{1+x} (1) \quad \div \quad 2x$$

$$= -x \cdot \sin x + \cos x - \frac{1}{1+x} \quad \div \quad 2x$$

$$= (1+x)^{-1}$$

$$= + (1+x)^{-2}$$

$$\frac{dy}{dx} = - (x \cdot \cos x + \sin x) - \sin x + \frac{1}{(1+x)^2} \quad \div \quad 2$$

$$= \frac{1}{(1+x)^2}$$

$$= \frac{1}{2}$$

$$b) \lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^3}$$

?

$$c) \lim_{x \rightarrow a} \frac{\ln(x-a)}{\ln(e^x - e^a)}$$

$$= \frac{\ln(a-a)}{\ln(e^a - e^a)} = \frac{\ln(0)}{\ln(0)} = \frac{-\infty}{-\infty} = x$$

$$\frac{dy}{dx} = \frac{\ln(x-a)}{\ln(e^x - e^a)}$$

$$= \frac{1}{x-a} \cdot (1-0) \quad \div \quad \frac{1}{e^x - e^a} (e^x - 0)$$

$$= \frac{1}{x-a} \quad \div \quad \frac{e^x}{e^x - e^a}$$

$$= \frac{e^x - e^a}{e^x (x-a)}$$

$$= \frac{e^x - e^a}{e^x x - a e^x}$$

$$= \frac{e^x - 0}{e^x - 0}$$

$$(e^x \cdot 1) + x \cdot e^x - a e^x$$

$$= \frac{e^x}{e^x + x \cdot e^x - a e^x}$$

$$= \frac{e^x}{e^x (1 + x - a)}$$

$$= \frac{1}{1 + x - a}$$

$$= \frac{1}{1 + a - a}$$

$$= \frac{1}{1}$$

$$= 1$$

$$= \frac{1}{1} = 1$$

$$e) \quad \lim_{x \rightarrow 0} \frac{\ln(\tan x)}{\ln(x)}$$

$$= \ln(\tan x) \div \ln(x)$$

$$\frac{d}{dx} = \frac{1}{\tan x} \times \sec^2 x \div \frac{1}{x}$$

$$= \frac{\sec^2 x}{\tan x} \times x$$

$$= \frac{x \cdot \sec^2 x}{\tan x}$$

$$\frac{d^2}{dx^2} = \frac{x \cdot 2\sec^2 x \tan x + \sec^2 x (x)}{\sec^2 x}$$

$$= \frac{\cancel{\sec^2 x} (x + \tan x + 1)}{\cancel{\sec^2 x}}$$

$$= 0 + 1$$

$$= 1$$